Ji 2020 cutaneous SCC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

8/10 samples (≥103 cells; data-derived gap 40→268, r=6.7) intermix on one continuous manifold — 0 detached islands (UMAP radius-graph). Cell cycle (35%) is an orthogonal compartment, not a separate lineage.

Gate 2 — module geography → GO (non-branching)

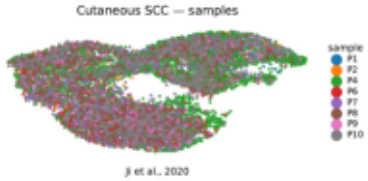
Gast/pSom co-vary (r=+0.11) and decline along pseudotime (ρ=−0.48 / −0.44); Neural Crest anti-correlated (r=−0.69 / −0.63), rising to a terminal readout (ρ=+0.35; crest-high decile mean pt=0.72). One dominant signature axis (67% of variance).

Gate 3 — hub mutual exclusivity → NA

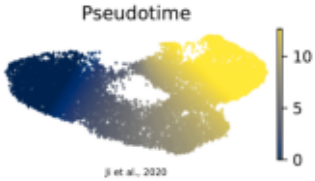
Single monotone axis — every trajectory tip is pSom-leaning (no Gast-dominant arm), so there is no bifurcation and no Mixed hub; no branch pair to test for mutual exclusivity.

STEP FIGURES

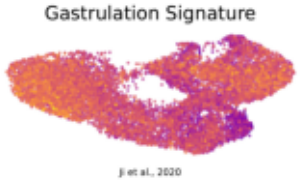
Manifold — samples



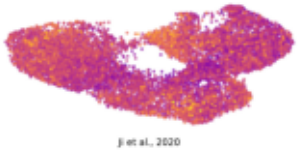
Pseudotime



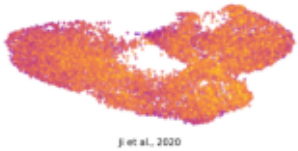
Gastrulation



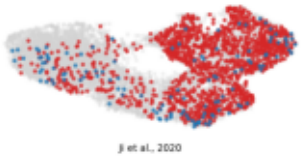
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Cutaneous SCC (Ji et al., 2020), 10x — 10 patients / 48,164 cells (atlas); malignant only: 10,598 loaded, 10,531 across 8 samples (≥103) enter the trajectory. Non-branching Linear, Gast/pSom → terminal Neural Crest.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Ji 2020's malignant manifold is a single monotone Gast/pSom → terminal-Crest gradient (linear), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. Cell cycle is orthogonal to the embryonic axis (pt ρ≈0), so the whole manifold is kept — the reference relationship holds in every compartment. EMT-TFs FOXC1 (early/root) → SNAI2 (late/terminus); canonical SNAI1/ZEB1/TWIST near the 10x detection floor.

Parameter choices

Min cells/samp.	≥103 (8/10; gap 40→268, r6.7)	**Integration**	Harmony(sample) θ=5; PCA 20PC
Root node	tip 2 = Gast/pSom-high, Crest-low	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Scoping	none — whole manifold	**Geometry**	Linear (non-branching); no hub
Subtypes	Ji 2020 calls + markers	**EMT-TFs**	FOXC1 (early) → SNAI2 (late)